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Machine learning reveals complex genetics of fungal resistance in sorghum grain mold

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Plant disease resistance is often a complex, polygenic trait, making its genetic dissection with traditional genome-wide association studies (GWAS) challenging. Grain mold in sorghum, a devastating disease caused by a fungal complex, exemplifies this complexity. We hypothesized that a machine learning (ML)-driven GWAS, employing diverse phenotypic representations from a panel of 306 sorghum accessions, could more effectively unravel the genetic basis of resistance. Phenotypic data, including raw disease scores, a 'difference phenotype' (inoculated vs. control), and principal components, were analyzed using Boosted Tree and Bootstrap Forest models, demonstrating strong explanatory power for phenotypic variance when trained on the entire dataset. This ML-GWAS approach confirmed a highly polygenic architecture for grain mold resistance, identifying numerous SNPs across the sorghum genome. Notably, several SNPs were consistently associated with resistance across multiple analytical models and phenotypic representations. These robustly identified SNPs were frequently located near genes with predicted functions integral to plant defense. Gene ontology (GO) analyses of the candidate gene set confirmed enrichment in categories supporting roles in pathogen recognition, DNA repair, and stress response modulation, indicating a multifaceted defense mechanism. This study provides valuable candidate genes for breeding sorghum with enhanced grain mold resistance and offers a refined methodological framework for dissecting complex traits in this crop. The successful application of this ML-based strategy in sorghum suggests its potential utility for studying similar complex traits in other plant species.

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INTRODUCTION

Understanding the genetic basis of complex traits, particularly those related to disease resistance in plants, is a central challenge in both evolutionary biology and agricultural science (Lynch and Walsh 1998; Gould 2010). Crop production worldwide is significantly impacted by diseases, with an estimated at least 10% of global food production lost annually to plant pathogens, exacerbating food insecurity for over 800 million people (Strange and Scott 2005; Burdon et al. 2006). Identifying the genes that control resistance is crucial for developing durable and sustainable agricultural practices and mitigating these losses. Sorghum [*Sorghum bicolor* (L.) Moench], a globally important cereal crop, provides a valuable model system for studying the complex genetics of disease resistance. While ranking fifth in world production and serving as a staple food for millions, particularly in arid and semi-arid regions (Frederiksen 1986), sorghum's versatility also extends to animal feed, bioenergy production, and various industrial applications (Frederiksen 1986; Ashok Kumar et al. 2011; Gonzalez et al. 2011). However, sorghum production is severely constrained by grain mold, a complex disease caused by a diverse array of fungal pathogens (Nida et al. 2019). These pathogens not only reduce yield and quality but also produce mycotoxins that pose significant risks to human and animal health (Sashidhar et al. 1992; Leslie et al. 2005; Oliveira et al. 2017; Ackerman et al. 2021). The complexity and variability of pathogens like those causing grain mold, combined with the genetic uniformity of many modern crop varieties resulting from domestication and selective breeding (Zohary 2004), creates a constant threat of devastating disease outbreaks, highlighting the urgent need for a deeper understanding of plant-pathogen interactions (Strange and Scott 2005). The sorghum grain mold disease complex is, therefore, an important target for genetic improvement.

Plant pathogens, however, are not merely agricultural concerns; they are also powerful evolutionary forces shaping the structure and dynamics of natural plant communities, from individual species to entire ecosystems (Burdon et al. 2006). Pathogen impacts range from subtle, largely unrecognized effects on individual plant fitness to catastrophic outbreaks that restructure entire communities. While the role of pathogens in natural ecosystems has often been underestimated, their influence is increasingly evident, particularly in the context of invasive species

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and ongoing environmental shifts (Burdon et al. 2006). The loss of biodiversity, driven by habitat destruction, agricultural intensification, and other human activities, disrupts natural ecosystem balances and can increase the risk of disease emergence and outbreaks (Maillard and Gonzalez 2006). Diverse ecosystems, with a wide range of plant species and genotypes, can provide a "buffer effect," limiting the spread of pathogens by reducing the density of susceptible hosts and increasing the probability that pathogens encounter resistant individuals (Maillard and Gonzalez 2006). Understanding how host genetic diversity, both within and between species, influences disease dynamics is therefore crucial for both conserving biodiversity and ensuring the long-term stability of agricultural systems.

To overcome these challenges, researchers have increasingly turned to exploring diverse germplasm collections, including landraces and improved lines, to identify novel sources of grain mold resistance (Nagesh Kumar et al. 2022). Institutions like the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) maintain extensive sorghum germplasm collections, representing a vast reservoir of genetic diversity (Sharma et al. 2010; Ashok Kumar et al. 2011). Subsets of this diversity, such as the Sorghum Association Panel (SAP), have been extensively studied to accelerate gene discovery (Casa et al. 2008; Rajan et al. 2012; Upadhyaya et al. 2013; Ahn et al. 2019, 2023). A powerful tool for connecting this genetic diversity to complex traits like disease resistance is the Genome-Wide Association Study (GWAS). GWAS uses statistical associations between genetic markers spanning the entire genome and variation in a trait of interest to pinpoint genomic regions likely to contain causal genes (Brachi et al. 2011; Korte and Farlow 2013). However, traditional GWAS methods, often relying on single-marker analysis and linear models, can struggle to fully capture the complex genetic architecture of traits like grain mold resistance, where numerous genes with small, potentially interacting effects, and significant genotype-by-environment interactions are involved (Mackay et al. 2009). The genetic architecture of these GxE interactions is often complex and polygenic (Boye et al. 2024), and environmental GWAS approaches in crops have indeed revealed multiple genomic regions contributing to such environment-dependent trait variations (Li et al. 2021). This is further complicated by the potential for non-genetic factors, such as variation in plant vigor or microenvironment, to influence observed disease severity, highlighting the need for approaches that can account for such confounding effects, for example, through the use of "difference phenotypes" that compare inoculated and control plants (Prom et al. 2021). This necessitates the exploration of more sophisticated analytical techniques.

Previous studies have utilized GWAS to identify genomic regions associated with grain mold resistance in sorghum, providing valuable initial insights (Cuevas et al. 2019; Nida et al. 2019; Prom et al. 2020). However, as discussed above, the highly polygenic nature of grain mold resistance, coupled with significant environmental influences and the complexity of the pathogen complex, likely limits the effectiveness of traditional GWAS approaches that primarily focus on single-marker associations. To address these limitations, ML methods offer a powerful alternative. Unlike traditional linear models, ML algorithms, such as Boosted Trees and Bootstrap Forests, can model complex, non-linear relationships between multiple genetic variants and phenotypic outcomes, including gene-gene interactions and gene-environment interactions (Bureau et al. 2005; Rigatti 2017; Basu et al. 2018). Specifically, Boosted Trees and Bootstrap Forests are ensemble methods that combine multiple decision trees to improve predictive accuracy and reduce overfitting (Breiman 2001; Bureau et al. 2005; Manual 2013; Chen 2014). These methods are well-suited to handling the high-dimensionality and complex interactions inherent in GWAS data, where the number of genetic markers far exceeds the number of individuals (Hoggart et al. 2008; Manual 2013; Sham and Purcell 2014).

The present study builds upon prior investigations that utilized various phenotypic approaches for sorghum grain mold (Prom et al. 2020, 2021) and aims to comprehensively analyze its genetic architecture using ML-driven GWAS. We hypothesized that applying ML models to a diverse set of phenotypic representations would yield a more nuanced understanding of the genetic basis of resistance than previously achieved. Furthermore, we anticipated that specific single-nucleotide polymorphism (SNP) markers would exhibit consistent associations across multiple treatments, models, and phenotypic representations, highlighting them as key contributors to resistance.

To address these hypotheses, our initial step involved performing ML-based genome-wide association analyses. These analyses utilized a range of phenotypic inputs, including raw treatment phenotypes (inoculation with *Alternaria alternata* [A], a mixture of *A. alternata, Fusarium thapsinum*, and *Curvularia lunata* [M], and an uninoculated control [C]), difference phenotypes (A-C and M-C), a combined phenotype ($A \times M \times C$), and the first principal component (PC1) of the phenotypic data, to capture both overall resistance levels and specific responses to pathogen inoculation. Following this, our second objective was to identify candidate genes located near the most significant and consistently associated SNPs, with a particular focus on those having potential roles in plant defense mechanisms. To enhance the robustness of these findings and provide a broader biological context, we aimed to compare the results from our ML models (Bootstrap Forest and Boosted Tree) across different phenotypic representations and also with findings from a traditional Mixed Linear Model (MLM) GWAS. This comparative approach was designed to pinpoint SNPs consistently identified across multiple analytical frameworks. Furthermore, GO enrichment analysis was planned for the candidate genes within these robustly identified regions to explore underlying biological pathways. These prioritized SNPs and associated enriched pathways are considered strong candidates for subsequent investigations into the molecular mechanisms of grain mold resistance.

MATERIALS AND METHODS

Phenotypic data

Phenotypic data for grain mold resistance were sourced from a previous study by Prom et al. who evaluated a larger set of 377 SAP lines (Prom et al. 2020). In brief, the SAP lines were evaluated for their response to inoculations A, M, and C. Field trials were conducted in Burleson County, TX, during the 2010 and 2013–2015 growing seasons using a randomized complete block design with three replications (Prom et al. 2020). Conidial suspensions of *A. alternata*, *F. thapsinum*, and *C. lunata* were prepared and adjusted to final concentrations of 2×10^4 conidia/ml for *A. alternata* and *C. lunata* and 1×10^6 conidia/ml for *F. thapsinum*. At 50% bloom, three panicles per accession/replicate were inoculated with 5–7 ml of spore suspension using a hand-held spray bottle. Control panicles were sprayed with sterile distilled water (Prom et al. 2020). Grain mold severity was assessed at maturity using a 1–5 scale as described by Thakur et al. (Thakur et al. 2006) and Isakeit et al. (2007), where 1 represents no mold, and 5 represents severe molding (≥50% of seeds affected). Histograms showing the distribution of these disease scores for treatments A, M, and C across the 306 sorghum accessions are provided in Supplementary Figs. S1–S3, respectively.

Genotypic data, genetic distance analysis, and correlation analysis

Genotypic data were obtained from a publicly available genotype-by-sequencing dataset for the SAP lines (Morris et al. 2013), which is composed of over 79,000 SNP markers. It is important to note that SNP data were only available for 306 of 377 SAP lines. To minimize potential false associations, SNPs with >15% missing data and minor allele frequency (MAF) < 5% were removed (Tabangin et al. 2009). Genetic distances between each pair of accessions were calculated using the Identity by State (IBS) method implemented in TASSEL 5 software (Bradbury et al. 2007). IBS estimates the proportion of alleles shared between individuals across all marker loci, providing a measure of genetic similarity (Stevens et al. 2011). Pearson's correlation coefficients were calculated using JMP Pro 17 software (SAS Institute Inc., Cary, NC) to assess the relationship between genetic distance (IBS) and phenotypic traits (grain mold scores for inoculations A, M, C, A-C, and M-C) (Klimberg 2023).

Population structure analysis

To explore population structure, hierarchical clustering was performed using both phenotypic and genotypic data in JMP Pro 17. For phenotype-based clustering, we used grain mold scores for the three treatments (A, M, and C). For genotype-based clustering, the analysis was based on the filtered SNP dataset. For both types of clustering, a distance matrix was calculated using the Ward method (Murtagh and Legendre 2014).To further visualize the relationships among accessions based on their phenotypic profiles, a Principal Component Analysis (PCA) was performed on the grain mold scores from treatments A, M, and C using JMP Pro 17. The first two principal components were then used to visualize the phenotypic structure.

Genome-wide association study using machine learning

To explore potentially complex, non-linear, or epistatic contributions of SNPs to phenotypic variation in grain mold severity, which might be overlooked by traditional single-marker association tests, an ML approach was employed. Given the high dimensionality of the genotypic data ($p = 63{,}442$ SNPs) and with the primary goal being the identification of potentially relevant SNPs based on their explanatory power within this dataset rather than optimizing predictive accuracy on unseen data, specific ML algorithms were selected and evaluated. The analysis was performed separately for each phenotypic trait: reactions to treatments A, M, and C, as well as the derived variations A-C, M-C, the interaction $A \times M \times C$, and Principal Component 1 (PC1, calculated from A, M, and C using JMP Pro 17).

To identify the most effective ML models for this analysis, we conducted a model screening process in JMP Pro 17 with default settings, comparing five different algorithms: Decision Tree, a model that creates a tree-like structure of prediction rules, (Song and Ying 2015); Bootstrap Forest (an adaption of Random Forest), an ensemble of decision trees built on bootstrapped data samples for improved accuracy and robustness, (Breiman 2001); Boosted Tree, an ensemble method that builds trees sequentially, each correcting its predecessor's errors, (Chen 2014); Boosted Neural Network, an approach combining multiple neural networks, trained sequentially to enhance overall performance, (Schwenk and Bengio 2000); and Support Vector Machine with radial basis function (RBF) kernel, a model that finds an optimal hyperplane for classification or regression, using the RBF kernel to handle non-linear data patterns, (Liu et al. 2011). This initial screening process, considering their ability to model complex relationships in the training data and their inherent suitability for variable importance estimation from high-dimensional genomic data, highlighted Bootstrap Forest and Boosted Tree as promising candidates for our ML-GWAS approach.

To further assess model generalization performance and evaluate predictive accuracy for these models, a 5-fold cross-validation ($K = 5$) scheme was implemented within the JMP Pro 17 platform during this screening phase, using a fixed random seed (1) for reproducibility. This cross-validation revealed that while models often achieved high explanatory power on the training sets (R squared values for Bootstrap Forest and Boosted Tree were frequently above 0.80 and ~0.70, respectively, across phenotypes when subsequently trained on the full dataset), their performance on the validation sets was highly variable and often poor, with R squared values ranging from negative values to below 0.3 across the evaluated models and phenotypes.

Consequently, for the primary ML-GWAS aimed at identifying genomic regions associated with phenotypic variation, Bootstrap Forest and Boosted Tree were selected as the primary models. This decision was based on several considerations: given the unreliability of validation set performance for robust model assessment or SNP importance estimation in this specific high-dimensional, limited-sample context, the main objective was to leverage the entire dataset for the most stable estimation of SNP contributions. Furthermore, these two models were chosen due to their strong explanatory power observed during the initial screening (on training sets) and their inherent capabilities for estimating variable importance. Subsequently, for the ML-GWAS and SNP importance ranking, these two algorithms were trained using the entire available dataset for each phenotypic trait (reactions to treatments A, M, and C, as well as the variations A-C, M-C, $A \times M \times C$, and PC1). This approach aimed to maximize the information used for identifying SNPs that contribute to explaining the phenotypic variance within the studied population. For the Bootstrap Forest method, the settings were: number of trees in the forest = 100, number of terms sampled per split = 1, bootstrap sample rate = 1, minimum splits per tree = 10, maximum splits per tree = 2000, minimum size split = 5, and random seed = 1. For the Boosted Tree method, the settings were as follows: learning rate = 0.1, minimum size split = 5, maximum layers = 6, number of layers to estimate = 95, sampling rate = 1, and random seed = 1. The SNPs were then ranked based on their importance scores (portion) derived from these full-dataset models.

In parallel, and for comparative purposes, a traditional Genome-Wide Association Study was conducted using a MLM for each of the same phenotypic traits and the identical filtered SNP dataset. These MLM analyses were performed using TASSEL version 5.2.96 (Bradbury et al. 2007). The MLM accounted for population structure by incorporating the first five principal components derived from a PCA of the genotypic data as fixed effects and kinship among individuals using a kinship matrix (Centered_IBS). The top 100 SNPs based on p value were considered significant from the MLM analyses for subsequent comparison with the ML results.

Candidate detection

The importance of each SNP in explaining the variation in the trait of interest was assessed using the "Portion", which represents the relative contribution of each SNP to the overall model precision. The results are visualized in Manhattan plots, which display the portion scores. Candidate genes nearest to the top SNPs were identified using the *Sorghum bicolor* reference genome v3.1.1, accessed through Phytozome 13 (https://phytozome-next.jgi.doe.gov/) (Goodstein et al. 2012). To define the genomic region around significant SNPs for candidate gene searches, Linkage Disequilibrium (LD) analysis was performed. Pairwise LD (R squared) between SNPs of each prioritized SNP was calculated using TASSEL version 5.2.96 (Bradbury et al. 2007). LD blocks were defined as regions where SNPs exhibited an R squared ≥0.6 with the prioritized SNP. All genes residing within these defined LD blocks were considered potential candidates and are reported in Supplementary Data 1. For each phenotype-ML model combination, the top 100 SNPs with the highest portion scores were selected. Similarly, from the MLM analyses, the top 100 SNPs exhibiting the lowest p values for association with each phenotype were selected.

Gene ontology enrichment analysis

A GO enrichment analysis was conducted to investigate the potential biological functions and pathways overrepresented among the candidate genes associated with grain mold resistance. The input for this analysis consisted of a comprehensive list of unique gene identifiers. This list comprised all genes located within the LD blocks of prioritized SNPs identified by our Bootstrap Forest and Boosted Tree models (as detailed in Supplementary Data 1). This curated list of unique *S. bicolor* gene IDs was submitted to ShinyGO v0.82 (available at http://bioinformatics.sdstate.edu/go/) (Ge et al. 2020). The enrichment analysis was conducted using the *Sorghum bicolor STRING* database. Default settings were applied, including a False Discovery Rate (FDR) cutoff of 0.05 to determine statistical significance. Pathway size filters were set to a minimum of 2 and a maximum of 5000 genes per GO term.

RESULTS

Disparate genetic and phenotypic structures revealed by grain mold resistance analysis in sorghum

To investigate the relationship between genetic relatedness and phenotypic responses to grain mold in sorghum, we analyzed a panel of 306 diverse accessions. Phenotypic responses were assessed after inoculations A, M, and C. Fig. 1 presents the results of hierarchical clustering, PCA, and correlation analyses exploring the structure of the phenotypic and genotypic data. Two-way hierarchical clustering based on the phenotypic data revealed 16 distinct clusters among 306 accessions (Fig. 1a). Notably, the clustering grouped A and M treatments together, yet separate from the control, suggesting similar phenotypic responses to both inoculation treatments. To further visualize the relationships

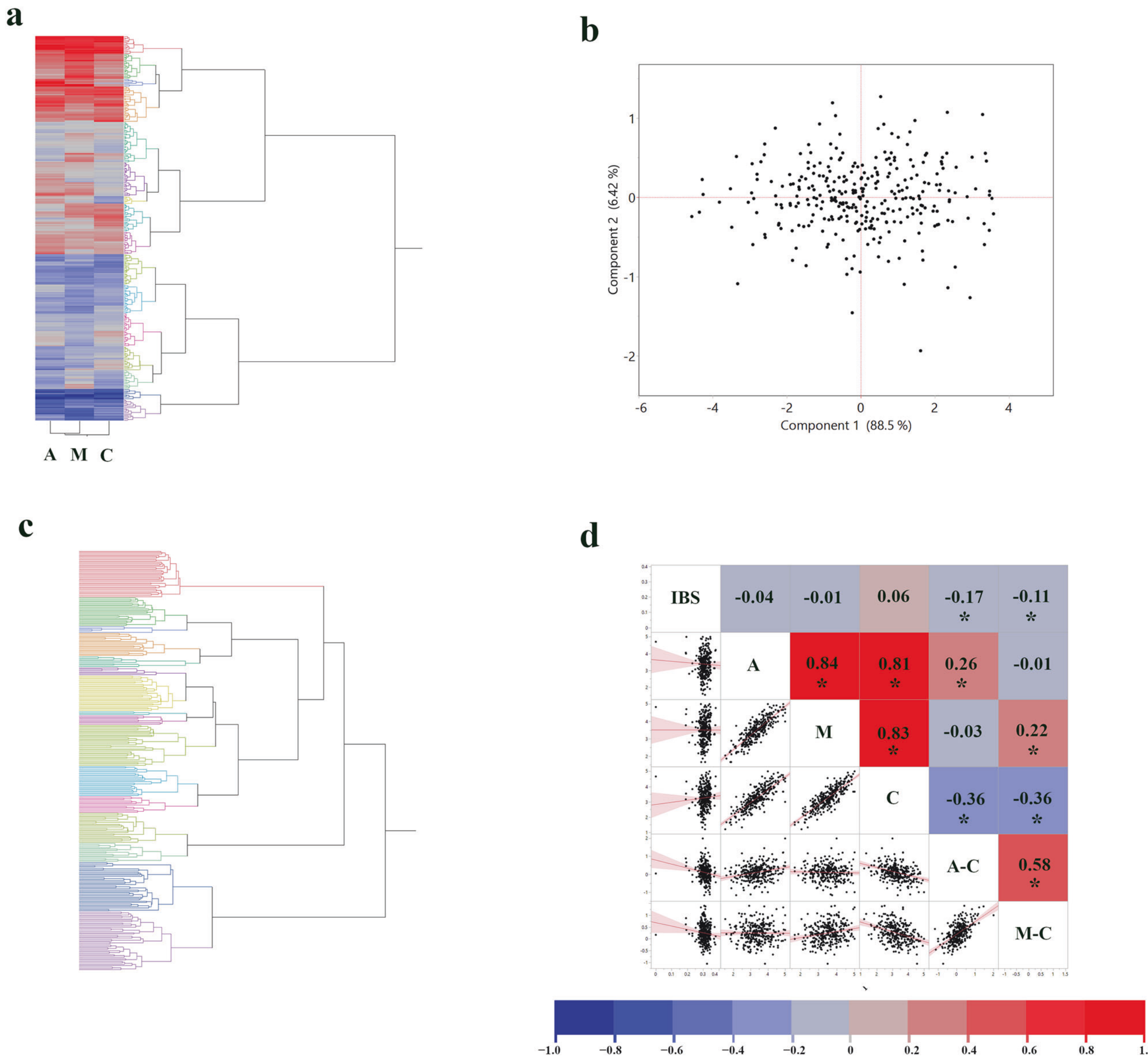

Fig. 1 Population structure and genetic-phenotypic correlations in sorghum grain mold resistance. **a** Hierarchical clustering of 306 sorghum accessions based on phenotypic responses to grain mold, including *A. alternata* (A), a mixture of *A. alternata*, *F. thapsinum*, and *C. lunata* (M), and control (C) treatments. Dendrogram branches are colored according to 16 distinct phenotypic clusters. **b** Principal Component Analysis visualization of the phenotypic data. The first principal component (PC1) explains 88.5% of the phenotypic variance, and the second principal component (PC2) explains 6.42%, revealing a continuous distribution. **c** Hierarchical clustering of the same accessions based on SNP genotype data, revealing 16 genotypic clusters **d** Pearson correlation matrix showing relationships between IBS genetic distances and phenotypic responses (A, M, C, A-C, and M-C). Asterisks (*) indicate significant correlations ($p < 0.05$).

among accessions based on their overall phenotypic profiles, a PCA was performed on the phenotypic data (Fig. 1b). The first principal component (PC1) accounted for 88.5% of the total phenotypic variance, while the second principal component (PC2) explained an additional 6.42%. The scatter distribution observed in the PCA plot, along with the lack of distinct groupings, suggests a complex, polygenic basis for grain mold resistance, with many small-effect loci potentially contributing to the observed phenotypic variation (Boyle et al. 2017). Hierarchical clustering based on SNP genotype data also identified 16 distinct clusters (Fig. 1c).

To further explore the relationship between genetic relatedness and phenotypic responses, we calculated the IBS genetic distance between all pairs of accessions. Pearson's correlation coefficients were then used to assess both the relationship between these genetic distances and the phenotypic responses to grain mold (Fig. 1d). The correlations between IBS and the three individual treatments (A, M, C) were weak, with the strongest correlation observed between IBS and A-C ($r = -0.17$, $p = 0.0035$) and between IBS and M-C ($r = -0.11$, $p = 0.046$). As such, genetic relatedness, as measured by the IBS method, appears to be a rather poor predictor of phenotypic similarity for grain mold resistance in this panel. However, genetically similar accessions seemed to exhibit somewhat more divergent responses between inoculation and control treatments. The phenotypic responses across treatments were highly correlated with each other. Specifically, the correlation between responses to treatments A and M was very high ($r = 0.84$, $p < 0.0001$), as were the correlations

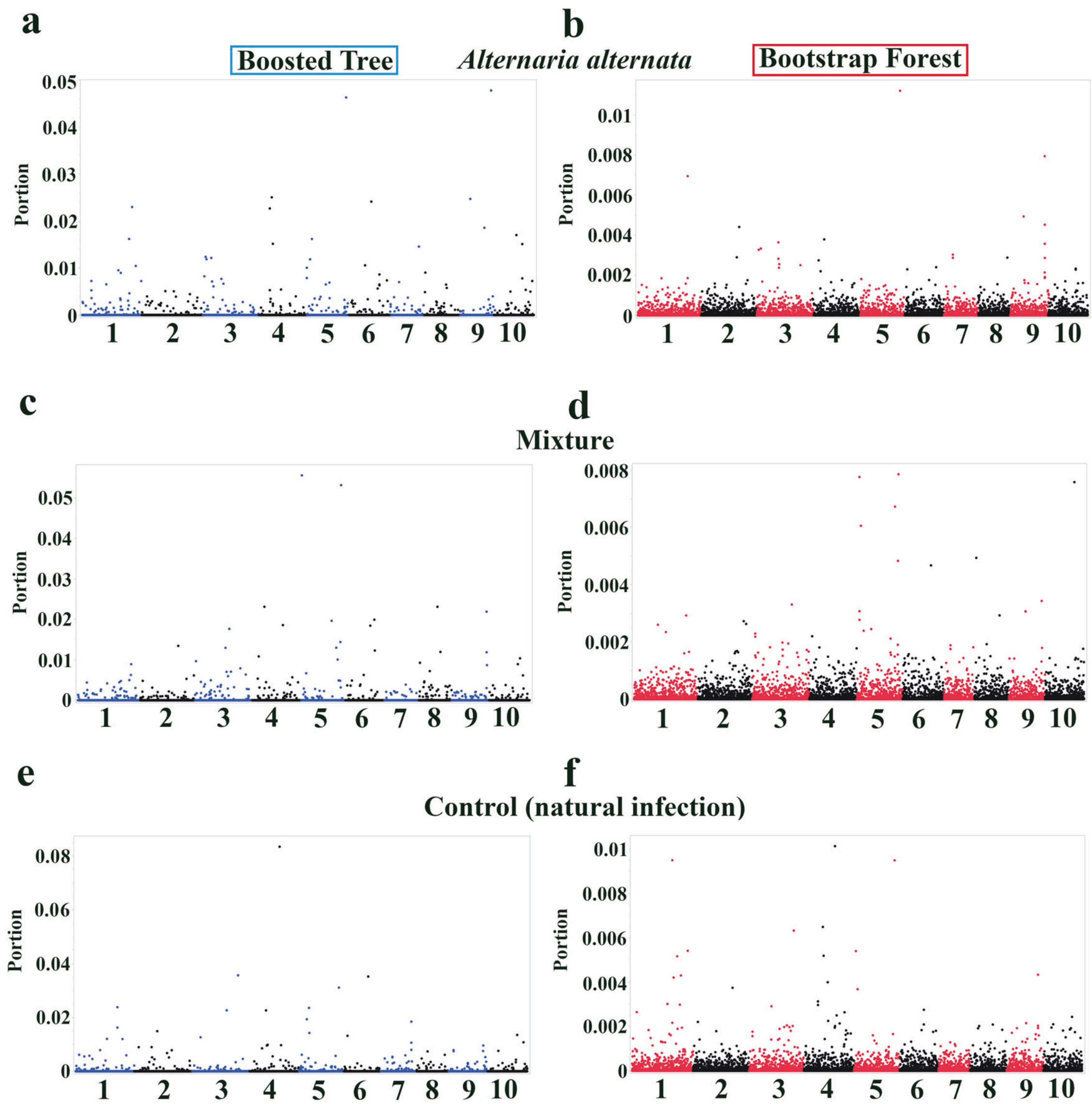

Fig. 2 Manhattan plots showing genome-wide association results for grain mold resistance in sorghum under different treatments. a *A. alternata* inoculation (Boosted Tree). **b** *A. alternata* inoculation (Bootstrap Forest). **c** Mixture inoculation (Boosted Tree). **d** Mixture inoculation (Bootstrap Forest). **e** Control (natural infection) (Boosted Tree). **f** Control (Bootstrap Forest). Each point represents a single SNP, with its position on the x-axis indicating the chromosome and its position along that chromosome. The y-axis represents the portion, a measure of variable importance derived from the respective ML models.

between responses to treatments C and A ($r = 0.81$, $p < 0.0001$) and treatments C and M ($r = 0.83$, $p < 0.0001$).

Identification of candidate SNPs and genes associated with grain mold resistance through machine learning models

Utilizing Boosted Tree and Bootstrap Forest models, a GWAS was conducted on 306 sorghum accessions to identify candidate SNPs and genes associated with grain mold resistance under three treatment conditions (inoculations A, M, and C, the last one representing natural infection). Manhattan plots in Fig. 2 visualize the GWAS results, displaying the top five SNPs contributing to grain mold resistance across the sorghum genome for each treatment and model. SNP importance was quantified using the portion. Notably, several SNPs on chromosomes 4, 5, and 9 consistently ranked high across multiple treatments and models, implying that these regions may harbor genes involved in broad-spectrum resistance to grain mold.

Uncovering SNPs linked to enhanced grain mold resistance via a treatment-control phenotype analysis

To further refine our analysis and identify genomic regions associated with the differential response to grain mold inoculation, we performed an ML-based GWAS using the difference between treatment and control conditions (A-C and M-C) as the phenotypic input. This approach aimed to highlight SNPs that are specifically associated with the effect of inoculation, potentially mitigating the influence of natural field infection (Prom et al. 2021). The same Boosted Tree and Bootstrap Forest ML models used in the previous analyses were employed for this analysis.

The results are presented in Fig. 3, which depicts Manhattan plots that display the importance of each SNP in explaining the variation in the A-C and M-C phenotypes for each model. Similar to the analysis of the raw treatment data (Fig. 2), several SNPs on chromosomes 2 and 5 exhibited relatively high importance scores across models and treatments (Fig. 3).

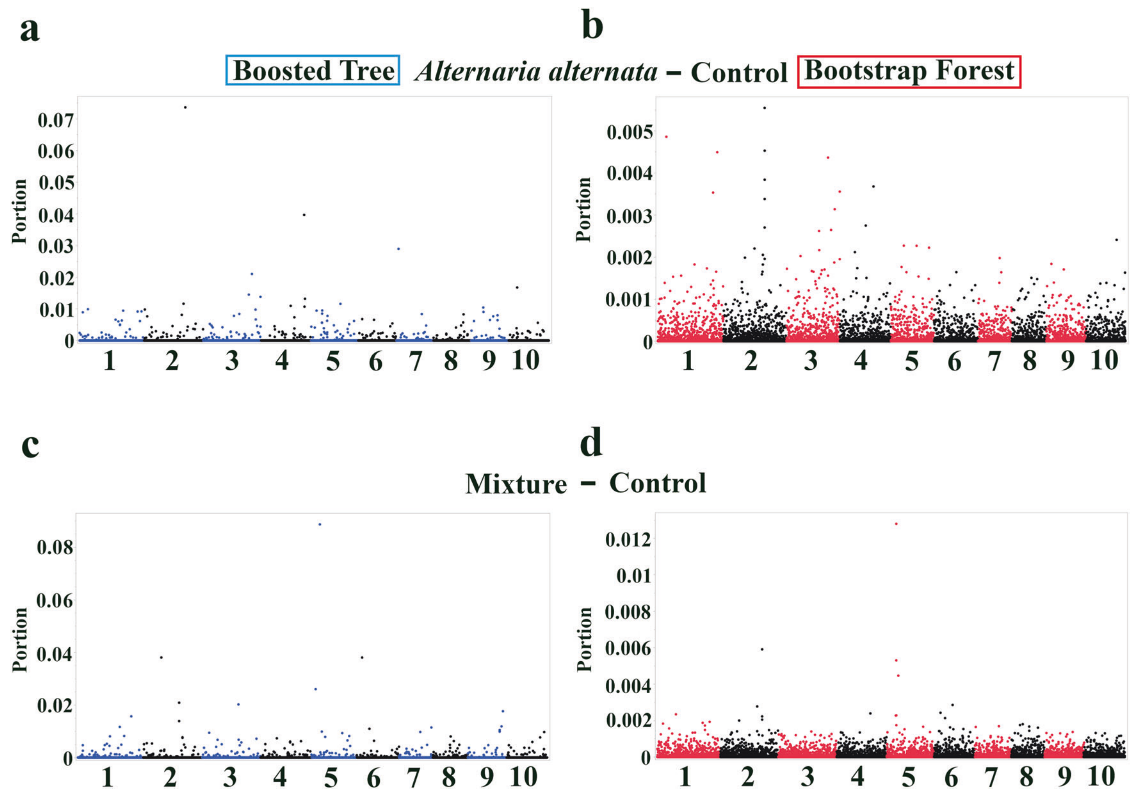

Fig. 3 Manhattan plots showing genome-wide association results for the difference in grain mold response between treatment and control conditions in sorghum SAP lines. a*A. alternata* inoculation minus control (A-C) (Boosted Tree). **b***A. alternata* inoculation minus control (A-C) (Bootstrap Forest). **c** Mixture inoculation minus control (M-C) (Boosted Tree). **d** Mixture inoculation minus control (M-C) (Bootstrap Forest). Each point represents a single SNP, with its position on the x-axis indicating the chromosome and its position along that chromosome. The y-axis represents the portion score.

Identification of candidate SNPs associated with a combined phenotype and principal component 1 of grain mold resistance

To explore alternative phenotypic representations of grain mold resistance and identify associated genomic regions, we performed a genome-wide association analysis using a combined phenotypic score (A × M × C) and the PC1 derived from responses to treatments A, M, and C. The combined phenotype, calculated as the product of the three treatment responses, aimed to capture a composite measure of overall resistance. PC1, on the other hand, represented the major axis of variation across the three treatment responses, potentially reflecting a more general resistance mechanism.

The results of the GWAS using the combined phenotype and PC1 are shown in Fig. 4, where Manhattan plots display the importance (portion) of each SNP in explaining these two phenotypes for both the Boosted Tree and Bootstrap Forest models. Similar to the analyses of individual treatments and treatment differences (Figs. 2 and 3), the Manhattan plots for the combined phenotype and PC1 showcase a complex landscape of SNP associations, with multiple SNPs across different chromosomes exhibiting relatively high importance scores (Fig. 4). This pattern reinforces the polygenic nature of grain mold resistance and the theory that multiple genomic regions contribute to the observed phenotypic variation.

While some SNPs were uniquely associated with individual treatments or treatment differences, others appeared repeatedly across multiple analyses (Tables 1 and 2), hinting at their potential involvement in broader resistance mechanisms. For instance, SNP S3_65427579 was identified in all four analyses presented in Table S1. Other frequently occurring SNPs include S9_57057698 (near *Sobic.009G230300*; uncharacterized protein, but the LD block contains Myb/SANT-like DNA-binding domain-containing protein and F-box), S9_11977630 (near *Sobic.009G082400*; uncharacterized protein, and the LD block contains numerous protein kinases), S9_57081206 (*Sobic.009G230700*; Myb/SANT-like DNA-binding domain-containing protein), S5_1329789 (*Sobic.005G014700*; cysteine desulfurase), S4_11186263 (*Sobic.004G113400*; cyclic nucleotide-binding domain-containing protein), S4_54114630 (near *Sobic.004G189301*; uncharacterized protein/F-box domain and zinc finger containing proteins in the LD block), S5_7154499 (near *Sobic.005G064500*; uncharacterized protein/peptidase A1 domain-containing protein in the LD block), and S2_72857837 (*Sobic.002G369301*; uncharacterized protein). Notably, SNP S5_60278659 (*Sobic.005G141700*; uncharacterized protein) appeared 8 times across all analyses, and *Sobic.002G270800* (PROTEIN ROOT PRIMORDIUM DEFECTIVE 1) was associated with four SNPs in three analyses. These repeatedly identified SNPs represent high-confidence candidates for further investigation, as their consistent association with grain mold resistance across different models, treatments, and phenotypic representations may stem from potentially robust and biologically relevant roles in the underlying genetic mechanisms.

Overlap among top candidate SNPs identified by different GWAS methodologies

To enhance confidence in candidate SNPs associated with grain mold resistance, we compared the top 100 SNPs identified by the traditional MLM with those from the Bootstrap Forest and Boosted Tree models. This comparison was performed for each analyzed trait: A (inoculation with *A. alternata*), C (control), M (mixed fungal inoculation), the difference phenotypes A-C and M-C, and the PC1. The extent of overlap among these top SNP sets is illustrated in Fig. 5.

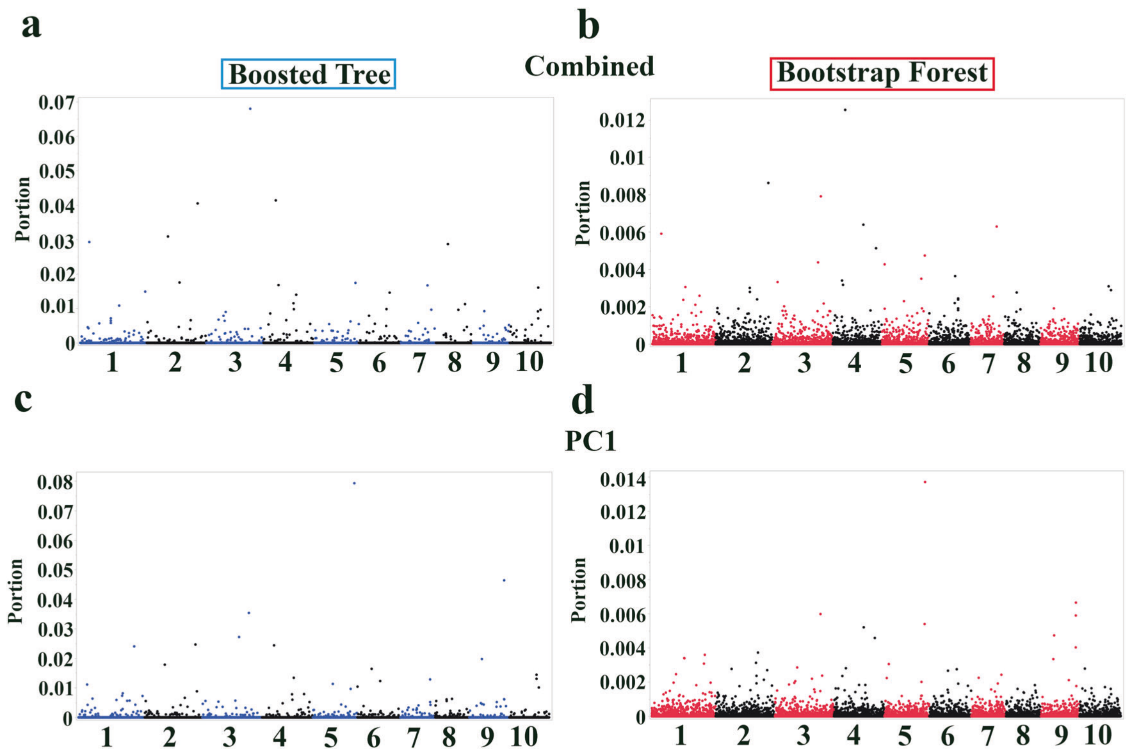

Fig. 4 Genome-wide association results for combined phenotypic score and PC 1 in sorghum grain mold resistance. Manhattan plots showing machine learning-based GWAS results for **a**, **b** a combined phenotypic score (A × M × C) and **c**, **d** the first principal component (PC1) derived from grain mold response data in sorghum. Both A × M × C and PC1 are derived from the responses of treatments A, M, and C. Models used were **a**, **c** Boosted Tree and **b**, **d** Bootstrap Forest. Each point represents a single SNP, with its position on the x-axis indicating the chromosome and its position along that chromosome. The y-axis represents the portion.

For Treatment A (Fig. 5a), three SNPs were consistently identified within the top 100 by all three methods: the newly prioritized S4_16290915 (near *Sobic.004G128666*, an uncharacterized protein, and also near an F-box domain gene, Supplementary Data 1), S9_57057698 (near *Sobic.009G230300*, uncharacterized protein, Myb/SANT-like DNA-binding domain-containing protein & F-box in the LD block), and S5_60278659 (near *Sobic.005G141700*, uncharacterized protein). Beyond this three-way consensus, Bootstrap Forest and Boosted Tree shared 16 SNPs. In comparison, MLM shared 2 SNPs with Bootstrap Forest and 2 with Boosted Tree, with 79, 79, and 93 SNPs being unique to Bootstrap Forest, Boosted Tree, and MLM, respectively.

In the analysis of Treatment C (Fig. 5b), four SNPs emerged as common to the top 100 lists of MLM, Bootstrap Forest, and Boosted Tree: S4_11186263 (near *Sobic.004G113400*, a cyclic nucleotide-binding domain-containing protein), S5_60278659 (near *Sobic.005G141700*, uncharacterized protein), the newly listed S7_62964494 (on the coding sequence of *Sobic.007G198200*, an uncharacterized protein with no other genes noted in its LD block), and S4_54114630 (near *Sobic.004G189301*, uncharacterized protein). Further pairwise overlaps included 17 SNPs between Bootstrap Forest and Boosted Tree, 6 between MLM and Boosted Tree, and 2 between MLM and Bootstrap Forest. Unique top SNPs numbered 77 for Bootstrap Forest, 73 for Boosted Tree, and 88 for MLM.

For Treatment M (Fig. 5c), five SNPs demonstrated consistently high importance or significance across all three methodologies. These included S4_11186263 (near *Sobic.004G113400*, cyclic nucleotide-binding domain-containing protein), S5_60278659 (near *Sobic.005G141700*, uncharacterized protein), S5_58050166 (near *Sobic.005G133900*, succinate dehydrogenase assembly factor 2, mitochondrial), S5_1329789 (near *Sobic.005G014700*, cysteine desulfurase), and S9_57081206 (near *Sobic.009G230700*, Myb/SANT-like DNA-binding domain-containing protein). An additional 16 SNPs were shared between Bootstrap Forest and Boosted Tree, while MLM shared 5 SNPs each with Boosted Tree and Bootstrap Forest. The counts of unique SNPs for Bootstrap Forest, Boosted Tree, and MLM were 74, 74, and 85, respectively.

The comparison for the A-C difference phenotype (Fig. 5d) revealed no SNPs common to all three analytical methods. Only a single SNP was shared between the top 100 lists of Bootstrap Forest and MLM, indicating that for this phenotype, the top SNPs identified were largely unique to each method (92 for Boosted Tree, 99 for Bootstrap Forest, and 91 for MLM).

In contrast, for the M-C difference phenotype (Fig. 5e), two SNPs were robustly identified in the top 100 by MLM, Bootstrap Forest, and Boosted Tree. These were S5_7154499 (near *Sobic.005G064500*, uncharacterized protein) and the newly identified S2_52403691. The latter SNP is noteworthy as it resides in a small LD block with no immediately annotated gene but is situated between two genes of interest: *Sobic.002G167600* (cysteine dioxygenase) and *Sobic.002G167700* (RNase H type-1 domain-containing protein). Further pairwise overlaps for M-C involved 11 SNPs between Bootstrap Forest and Boosted Tree, 5 between MLM and Boosted Tree, and 4 between MLM and Bootstrap Forest, with 82, 83, and 89 SNPs unique to Boosted Tree, Bootstrap Forest, and MLM, respectively.

Finally, for the PC1 phenotype (Fig. 5f), five SNPs were consistently found in the top 100 across all three methods: S4_11186263 (near *Sobic.004G113400*, cyclic nucleotide-binding domain-containing protein), S5_60278659 (near *Sobic.005G141700*, uncharacterized protein), the newly prioritized S4_55321911 (on the coding sequence of *Sobic.004G201400*, a PGG domain-containing protein, with no strong LD noted with other genes), S5_58050166 (near *Sobic.005G133900*, succinate dehydrogenase assembly factor 2, mitochondrial), and S2_72857837 (near

Table 1. Top candidate SNPs associated with grain mold resistance in sorghum under different treatments, identified by Boosted Tree and Bootstrap Forest machine learning models.

SNP ID	Nearest gene and function	Base pairs away	Importance score (portion)/MLM based *p* value
Alternaria alternata—Boosted Tree			
S9_57057698	*Sobic.009G230300* Uncharacterized protein	0	0.048/0.00019
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.046/0.00017
S4_14032864	*Sobic.004G123801* Subtilisin-like protease fibronectin type-III domain-containing protein	22,662	0.025/0.028
S9_11977630	*Sobic.009G082400* Uncharacterized protein	12,948	0.025/0.0282
S6_48965156	*Sobic.006G124100* Glycosyltransferase	0	0.024/0.0015
Alternaria alternata—Bootstrap Forest			
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.011/0.00017
S9_57057698	*Sobic.009G230300* Uncharacterized protein	0	0.0079/0.00019
S1_63595560	*Sobic.001G346500* Vacuolar cation/proton exchanger	0	0.0069/0.0045
S9_11977630	*Sobic.009G082400* Uncharacterized protein	12,948	0.0049/0.0282
S9_57081206	*Sobic.009G230700* Myb/SANT-like DNA-binding domain-containing protein	4842	0.0045/0.00077
Mixture—Boosted Tree			
S5_1329789	*Sobic.005G014700* Cysteine desulfurase	0	0.056/0.001
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.053/0.000023
S8_43114582	*Sobic.008G093300* Methylthioribose-1-phosphate isomerase-like	124,761	0.023/0.013
S4_11186263	*Sobic.004G113400* Cyclic nucleotide-binding domain-containing protein	198	0.023/0.00094
S9_57081206	*Sobic.009G230700* Myb/SANT-like DNA-binding domain-containing protein	4842	0.022/0.0014
Mixture—Bootstrap Forest			
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.0079/0.000023
S5_1329789	*Sobic.005G014700* Cysteine desulfurase	0	0.0078/0.001
S10_52171186	*Sobic.010G182300* Elongation factor 1-alpha	27,498	0.0076/0.035
S5_58050166	*Sobic.005G133900* Succinate dehydrogenase assembly factor 2, mitochondrial	1718	0.0067/0.00062
S5_2229600	*Sobic.005G024700* Major facilitator superfamily (MFS) profile domain-containing protein	1434	0.0061/0.02
Control—Boosted Tree			
S4_54114630	*Sobic.004G189301* Uncharacterized protein	597	0.083/0.0022
S3_65427579	*Sobic.003G329100* Protein MIZU-KUSSEI 1	2663	0.036/0.35

Table 1. continued

SNP ID	Nearest gene and function	Base pairs away	Importance score (portion)/MLM based *p* value
S6_52764761	*Sobic.006G171200* Uncharacterized protein Similar to Expansin-B15 precursor	1055	0.035/0.0077
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.031/0.000024
S1_56528270	*Sobic.001G288700* Armadillo repeat-containing domain-containing protein	0	0.024/0.01
Control—Bootstrap Forest			
S4_54114630	*Sobic.004G189301* Uncharacterized protein	597	0.01/0.0022
S1_56528297	*Sobic.001G288700* Armadillo repeat-containing domain-containing protein	0	0.0095/0.01
S5_60278659	*Sobic.005G141700* Uncharacterized protein Similar to DNA topoisomerase, D-mannose binding lectin, and leucine-rich repeat	3147	0.0095/0.000024
S4_23004829	*Sobic.004G134300* DUF4283 domain-containing protein	121,047	0.0065/0.044
S3_65427579	*Sobic.003G329100* Protein MIZU-KUSSEI 1	2663	0.0063/0.35

SNPs are listed in descending order of importance based on the portion, a measure of the relative contribution of each SNP to the overall model accuracy. The table includes the SNP ID, nearest gene, putative function of the nearest gene, the distance between the SNP and the nearest gene, the importance score (portion) along with the MLM-based *p* value for the association of that SNP with the trait, the model in which the SNP was identified as important, and the treatment condition. Detailed information on LD blocks (R squared ≥ 0.6) surrounding these prioritized SNPs, including all genes within these blocks, is provided in Supplementary Data 1.

Sobic.002G369301, uncharacterized protein). Additional pairwise overlaps included 12 SNPs between Bootstrap Forest and Boosted Tree, 4 between MLM and Boosted Tree, and 2 between MLM and Bootstrap Forest. The number of SNPs uniquely identified by Bootstrap Forest, Boosted Tree, or MLM was 81, 79, and 89, respectively.

Functional annotation of candidate genes using GO enrichment analysis

A GO enrichment analysis was performed to explore the potential biological functions of the candidate genes associated with grain mold resistance. The analysis utilized a list of 327 unique candidate gene identifiers (located within the LD blocks of prioritized SNPs identified through our ML models; Supplementary Data 1). Of these, 244 genes were successfully mapped and recognized by the ShinyGO v0.82 database for *Sorghum bicolor STRING* database and were included in the enrichment analysis. The GO enrichment analysis revealed several significantly over-represented (FDR < 0.05) biological processes, molecular functions, and cellular components among the candidate genes (Fig. 6; Supplementary Data 1 for a full list of enriched terms). Notably, functions related to defense and stress responses, signaling, and DNA maintenance were prominent. Among the most highly significant and strongly enriched terms were those associated with specific protein domains and signaling components, such as “LRAT domain, and Protein kinase C conserved region 2 (CalB)” (FDR = 6.0E-03, Fold Enrichment = 84.8, 3 genes) and ‘Mixed, incl. zinc finger, fyve/phd-type’ (FDR = 6.0E-03, Fold Enrichment = 84.8, 3 genes). Several terms pointing towards immune system processes, analogous to those in animals, were also significantly enriched. These included “Response to type I interferon” (FDR = 4.3E-02, Fold Enrichment = 94.3, 2 genes), “Interferon-gamma-mediated signaling pathway” (FDR = 4.3E-02, Fold Enrichment = 94.3, 2 genes), and “Cellular response to interleukin-1” (FDR = 4.3 E-02, Fold Enrichment = 94.3, 2 genes). Furthermore, the analysis highlighted an overrepresentation of genes involved in DNA repair and genome stability, evidenced by the enrichment of terms like ‘FANCM-MHF complex’ (FDR = 4.3E-02, Fold Enrichment = 94.3, 2 genes) and “Fanconi anaemia nuclear complex” (FDR = 4.3E-02, Fold Enrichment = 70.7, 2 genes). Other enriched categories included molecular functions such as various oxidoreductase activities (e.g., “L-methionine-(s)-s-oxide reductase activity”, FDR = 4.3E-02, Fold Enrichment = 70.7, 2 genes) and broad terms like “DNA-binding” (FDR = 4.5E-02, Fold Enrichment = 2.6, 17 genes), suggesting a diversity of mechanisms potentially contributing to grain mold resistance.

DISCUSSION

This study demonstrates the power of ML-driven GWAS for dissecting the complex genetic architecture of a polygenic disease resistance trait in a model crop species. While grain mold resistance in sorghum has long been recognized as a complex, polygenic trait, with a continuous spectrum of phenotypic responses observed across diverse germplasm (Fig. 1b), traditional GWAS approaches have struggled to fully capture the underlying genetic complexity. The weak correlation between genetic distance and phenotypic similarity observed in our study, and in the evolution of complex traits generally (Boyd 2012), further supports the involvement of numerous genes with potentially small, additive, and interacting effects, as well as the influence of non-genetic factors. By employing ensemble ML methods (Boosted Trees and Bootstrap Forests) and integrating diverse phenotypic representations, we were able to identify a suite of candidate genes and genomic regions associated with grain mold resistance, many of which were not detected by previous studies using traditional GWAS approaches (Prom et al. 2020, 2021). These findings provide significant insights into the genetic architecture

Table 2. Top candidate SNPs associated with the differential response to grain mold (difference phenotypes A-C and M-C) in sorghum, identified by Boosted Tree and Bootstrap Forest machine learning models.

SNP ID	Nearest gene and function	Base pairs away	Importance score (portion)/MLM based *p* value
A-C—Boosted Tree			
S2_65438365	*Sobic.002G270800* PROTEIN ROOT PRIMORDIUM DEFECTIVE 1 (PRD1)	0	0.074/0.0086
S4_63380351	*Sobic.004G293700* Peroxisomal biogenesis factor 19	568	0.04/0.26
S7_75107	*Sobic.007G000500* HMG box domain-containing protein	0	0.029/0.17
S3_68275529	*Sobic.003G366500* Trafficking protein particle complex subunit	0	0.021/0.02
S10_7369881	*Sobic.010G085700* Uncharacterized protein	1501	0.017/0.000011
A-C—Bootstrap Forest			
S2_65438365 S2_65441818 S2_65441267	*Sobic.002G270800* PROTEIN ROOT PRIMORDIUM DEFECTIVE 1 (PRD1)	0	0.0055/0.0086
S1_7154618	*Sobic.001G092700* Uncharacterized protein	1340	0.0049/0.023
S1_69474797	*Sobic.001G412001* Uncharacterized protein	1722	0.0044/0.0023
S3_65724513	*Sobic.003G333100* Inositol polyphosphate-related phosphatase domain-containing protein	4842	0.0038/0.03
S4_56292309	*Sobic.004G213000* Tudor domain-containing protein	3382	0.0037/0.0092
M-C—Boosted Tree			
S5_7154499	*Sobic.005G064500* Uncharacterized protein	19,565	0.089/0.000079
S2_16115286	*Sobic.002G121900* K Homology domain-containing protein	0	0.038/0.89
S6_7023758	*Sobic.006G032300* HECT-type E3 ubiquitin transferase	8094	0.038/0.0015
S5_3774504	*Sobic.005G041100* BZIP domain-containing protein	2164	0.026/0.46
S2_61720117	*Sobic.002G225300* HSF-type DNA-binding domain-containing protein	7994	0.021/0.17
M-C—Bootstrap Forest			
S5_7154499	*Sobic.005G064500* Uncharacterized protein	19,565	0.013/0.000079
S2_65438715	*Sobic.002G270800* PROTEIN ROOT PRIMORDIUM DEFECTIVE 1 (PRD1)	0	0.0059/0.0024
S5_7106146	*Sobic.005G064300* Aspartyl protease	5031	0.0053/0.000053
S5_8996059	*Sobic.005G074200* Uncharacterized protein	400	0.0045/0.017
S6_46965889	*Sobic.006G099600* Ubiquitin-like protease family profile domain-containing protein	26,139	0.0029/0.0019

For each machine learning model and difference phenotype combination, the top five SNPs are listed in descending order of importance based on the portion. The table includes the SNP ID, nearest gene, putative function of the nearest gene, the distance between the SNP and the nearest gene, the importance score (portion), the model in which the SNP was identified as important, and the treatment condition (A-C or M-C). The top five candidate SNPs for each treatment and model combination are shown.

of grain mold resistance, offering valuable targets for breeding more resilient sorghum varieties.

A key methodological contribution of this study is the demonstration that ML-driven GWAS can effectively dissect complex traits even in the presence of significant environmental influences and correlations between experimental treatments. While we observed strong correlations among the different treatment conditions, including the untreated control, suggesting a substantial contribution of natural field infection, our ML approach, particularly the Bootstrap Forest model, demonstrated a high degree of explanatory power when trained on the full dataset for SNP identification (e.g., achieving an R^2 value of ~0.84

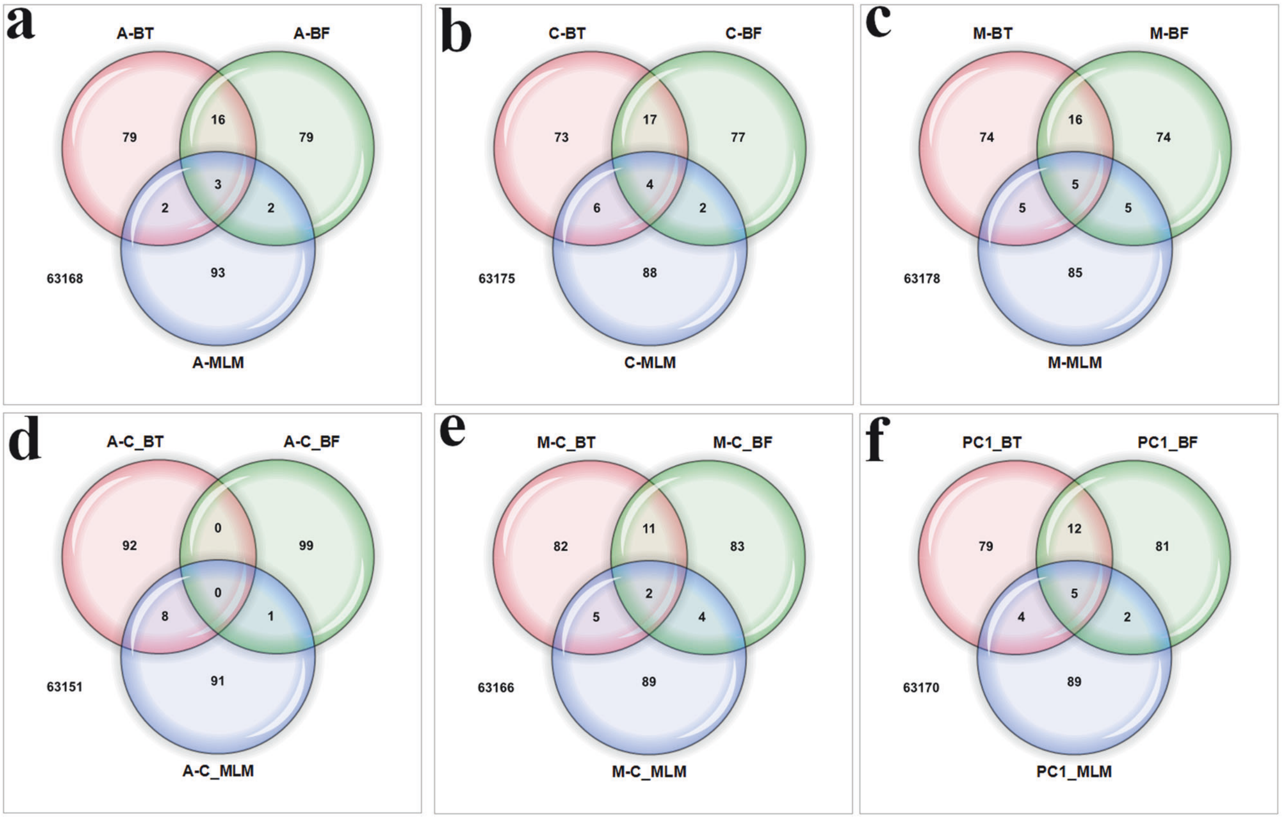

Fig. 5 Venn diagrams illustrating the overlap among the top 100 candidate SNPs identified by three different methodologies for various grain mold resistance phenotypes in 306 sorghum accessions. The methodologies compared are: Boosted Tree (BT, red circles), Bootstrap Forest (BF, green circles), and Mixed Linear Model (MLM, blue circles). Numbers within the diagrams indicate the count of SNPs unique to each method or shared within the intersecting regions for each specific phenotype. The phenotypes analyzed are: **a** Response to *A. alternata* inoculation (Treatment A). **b** Response under control conditions (Treatment C). **c** Response to mixed fungal inoculation (Treatment M). **d** Difference phenotype: *A. alternata* inoculation minus control (A-C). **e** Difference phenotype: Mixed fungal inoculation minus control (M-C). **f** First principal component (PC1) of phenotypic variance. The number outside and below each set of circles represents the total number of SNPs analyzed in the GWAS for that trait after quality filtering.

and an RMSE of 0.27 when modeling the *A. alternata* response within this dataset, data not shown). This effectiveness in capturing the complex phenotypic variance in the studied population can be attributed to the ability of ensemble methods like Boosted Trees and Bootstrap Forests to model complex, non-linear relationships between SNPs and phenotypes, including gene-gene and gene-environment interactions, which are likely prevalent in grain mold resistance (Boyle et al. 2017). Traditional GWAS, relying on single-marker linear regressions, often struggle to capture such interactions (Bureau et al. 2005; Rigatti 2017; Basu et al. 2018). Furthermore, our use of a “difference phenotype,” comparing inoculated and control plants, likely helped to mitigate the influence of non-genetic factors and natural infection, potentially revealing loci specifically associated with the response to inoculation that might have been masked when using only raw phenotypic scores. Indeed, comparisons of the GWAS results across different phenotypic representations revealed both overlapping and unique associations, suggesting that different aspects of the genetic architecture are captured by each approach. This multi-faceted analytical strategy, combining ML with diverse phenotypic representations, offers a powerful framework for dissecting complex traits in other plant species and beyond.

Among the candidate genes identified, *Sobic.005G141700*, *Sobic.003G329100*, and *Sobic.002G270800* stand out as particularly compelling due to their consistent association with grain mold resistance across multiple analyses and phenotypic representations. The most frequently identified SNP, S5_60278659, lies near *Sobic.005G141700*, an uncharacterized protein with predicted domains suggesting multifaceted roles in plant defense. These include a DNA topoisomerase domain, potentially involved in DNA repair and response to pathogen-induced stress (Choi et al. 2001; Sirikantaramas et al. 2008; Šimková et al. 2012); a D-mannose binding lectin domain, implicated in pathogen recognition and the initiation of defense signaling (Hwang and Hwang 2011; Sager et al. 2017; Barre et al. 2019); and a leucine-rich repeat (LRR) domain, a common feature of plant disease resistance (R) genes that mediate pathogen recognition (Jones and Jones 1997). This combination of domains suggests a potentially broad role for *Sobic.005G141700* in recognizing and responding to multiple grain mold pathogens. *Sobic.003G329100* encodes Protein MIZU-KUSSEI 1, previously implicated in plant-pathogen interactions, and *Sobic.002G270800* encodes PROTEIN ROOT PRIMORDIUM DEFECTIVE 1, which plays a role in mitochondrial function and stress responses (Wang et al. 2024). While our study did not reveal strong associations near the *MYB* transcription factors previously linked to flavonoid biosynthesis and grain mold resistance in Ethiopian sorghum landraces (Nida et al. 2019), the repeated identification of genes involved in diverse defense-related processes underscores the complex, polygenic nature of resistance to this disease complex.

The predicted functions of these consistently identified genes further support their potential roles in broad-spectrum disease resistance. The LRR domain of *Sobic.005G141700*, a hallmark of many plant R genes (Jones and Jones 1997), strongly suggests involvement in pathogen recognition, while the MIZU-KUSSEI 1 domain of *Sobic.003G329100* points towards a role in modulating

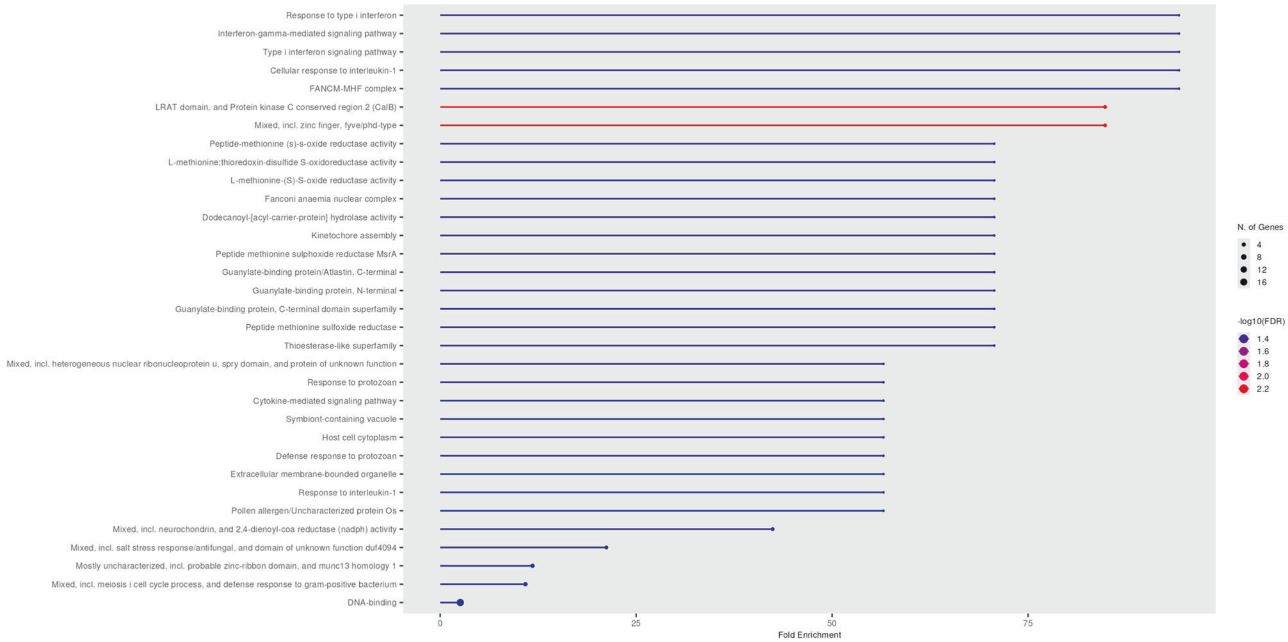

Fig. 6 GO enrichment analysis of candidate genes associated with grain mold resistance. The plot shows significantly enriched GO terms (FDR < 0.05). The x-axis represents fold enrichment. Dot color indicates the −log10(FDR) value (higher values/red colors indicate greater significance), and dot size corresponds to the number of input genes annotated to each term.

cellular responses to stress, as seen in other plant-pathogen interactions (Šubr et al. 2020). The predicted role of *Sobic.002G270800* in mitochondrial function (Wang et al. 2024) highlights the importance of cellular energy production and homeostasis in mounting an effective defense response. These diverse, yet interconnected, mechanisms, including pathogen recognition, stress response modulation, and mitochondrial function, suggest a complex interplay of pathways contributing to grain mold resistance in sorghum. This multifaceted resistance architecture, also observed in other studies of grain mold and other diseases (Cuevas and Prom 2024), may contribute to the durability of resistance by making it more difficult for pathogens to overcome multiple defense layers simultaneously. However, the ongoing coevolutionary arms race between plants and pathogens, as described by the Red Queen hypothesis (Van 1973), presents a constant challenge for breeding durable resistance. Pathogens are continually evolving to overcome plant defenses, making it crucial to identify and deploy a diverse range of resistance mechanisms. Further investigation into a broader set of candidate genes identified in this study and other works may reveal additional components of this complex defense network.

In addition to the three key genes highlighted above, our analyses identified several other candidate genes consistently associated with grain mold resistance across multiple phenotypic representations and models. These include genes encoding proteins with predicted functions ranging from Myb/SANT-like DNA-binding domains (*Sobic.009G230700*) and cyclic nucleotide-binding domains (*Sobic.004G113400*) to cysteine desulfurase activity (*Sobic.005G014700*), as well as several proteins of currently uncharacterized function (*Sobic.009G230300*, *Sobic.009G082400*, *Sobic.004G189301*, *Sobic.005G064500*, and *Sobic.002G369301*). While the function of the nearest gene awaits elucidation, the underlying causal variant might influence this uncharacterized gene or, importantly, other candidate genes residing within the same LD block (as detailed in Supplementary Data 1). Indeed, for many of the candidate genes identified in this study, their predicted domains and homology to genes involved in plant defense in other species suggest a broad range of potential mechanisms, including transcriptional regulation, signal transduction, and sulfur metabolism (Álvarez et al. 2012). This further reinforces the notion of a complex, multi-layered defense response, where multiple pathways likely contribute to overall resistance. It is also an excellent example of how genomic approaches can generate multiple, testable hypotheses regarding the mechanisms of plant defense. The identification of such a broad range of candidate genes with preliminary support for a wide variety of defense strategies further strengthens the case for potential durable resistance.

The identification of SNPs consistently ranked highly by these distinct analytical approaches, MLM's statistical tests versus the ML models' importance scores, lends greater confidence to these markers as being genuinely associated with grain mold resistance. These consensus SNPs, particularly those with plausible biological functions such as S5_1329789 (cysteine desulfurase for Treatment M) or those in intriguing genomic locations like S2_52403691 (between cysteine dioxygenase and RNase H genes for the M-C phenotype), represent high-priority candidates for further investigation and functional validation, particularly when associated with genes having plausible biological functions in plant defense. For instance, SNP S5_1329789, notably associated with Treatment M, is located near a gene annotated as a cysteine desulfurase. Enzymes of this type are crucial for plant immunity, playing roles in sulfur mobilization for the biosynthesis of iron-sulfur (Fe-S) clusters (essential cofactors for many defense-related proteins) and in the production of hydrogen sulfide (H_2S), an emerging signaling molecule in plant defense responses against pathogens (Fonseca et al. 2020; Choudhary et al. 2022). The broader context of cysteine metabolism and signaling is also increasingly recognized as vital for plant pathogen responses (Moormann et al. 2025). Other intriguing candidates include SNPs in specific genomic locations, such as S2_52403691 found between genes encoding a cysteine dioxygenase and an RNase H for the M-C phenotype. Such findings strongly warrant further investigation and functional validation. The observed variations in overlap across different traits and methods also suggest that while certain core genetic signals exhibit robustness, each analytical approach may also capture unique aspects of the complex genetic architecture underlying this trait.

Further supporting the involvement of these multifaceted defense mechanisms, our GO enrichment analysis of the candidate genes (derived from LD blocks of prioritized SNPs identified by our ML models) revealed a significant overrepresentation of several pertinent biological processes and molecular functions (Fig. 6). Notably, terms associated with DNA repair and genome integrity, such as "FANCM-MHF complex" (FDR = 4.3E-02) and "Fanconi anaemia nuclear complex" (FDR = 4.3E-02), were significantly enriched. The FANCM-MHF complex is known for its conserved role in DNA remodeling and maintaining genome stability under replication stress (Yan et al. 2010; Fox et al. 2014), and Fanconi anaemia nuclear complex is involved in the DNA repair pathway, a crucial network responsible for repairing DNA interstrand crosslinks and other complex lesions to maintain genome integrity (Walden and Deans 2014). The enrichment of these terms suggests that efficient DNA damage repair mechanisms are a key component of grain mold resistance in sorghum, likely aiding in coping with pathogen-induced cellular stress and DNA damage. This aligns with the predicted DNA topoisomerase domain in *Sobic.005G141700* and the general cellular stress response expected during pathogen interactions. Additionally, significant enrichment was observed for terms indicative of defense signaling and immune system processes. While pathways such as "Response to type I interferon" (FDR = 4.3E-02) and "Interferon-gamma-mediated signaling pathway" (FDR = 4.3E-02) are named based on vertebrate immunity, their enrichment in our plant gene set likely points to the involvement of analogous signaling cascades or conserved protein domains crucial for sorghum's defense. This interpretation is grounded in the recognized molecular similarities between plant and animal innate immune systems, which share common principles in pathogen recognition and downstream signaling components (Nürnberger et al. 2004). This is further supported by the enrichment of terms involving specific signaling domains like "LRAT domain, and Protein kinase C conserved region 2 (CalB)" (FDR = 6.0E-03). The overrepresentation of terms related to methionine metabolism, particularly "L-methionine-(s)-s-oxide reductase activity" (MSR; FDR = 4.3E-02), emphasizes the importance of mechanisms that repair oxidized methionine residues. Plant MSRs play crucial protective roles against oxidative damage to proteins during both abiotic and biotic environmental stresses, and are involved in redox homeostasis and signaling (Roy and Nandi 2017). Their enrichment in our candidate gene set suggests that maintaining protein integrity and function via methionine repair is a key component of sorghum's response to grain mold. This, coupled with the significant enrichment of broader functional categories like DNA-binding (FDR = 4.5E-02), indicative of the involvement of numerous transcription factors and other gene regulatory proteins known to orchestrate the extensive transcriptional reprogramming essential for plant immune responses (Pandey and Somssich 2009; Tsuda and Somssich 2015), further highlights the complex transcriptional and metabolic reprogramming central to developing grain mold resistance.

Building upon this foundation, it is crucial to acknowledge several limitations of our current study. While conducting our study in a single environment (Burleson County, TX) limits the generalizability of our findings to other regions with different climates and pathogen populations, it also provided a degree of experimental control that was beneficial for our primary goal: dissecting the complex genetic architecture of grain mold resistance under a defined set of conditions. By minimizing environmental variation, we aimed to more clearly isolate the genetic effects contributing to resistance and reduce the confounding influence of genotype-by-environment interactions, which are known to be significant for this trait (Thakur et al. 2006; Ashok Kumar et al. 2008). However, the identified SNPs and candidate genes may not be equally important, or may interact differently, in other environments or under different pathogen pressures. Furthermore, while the SAP represents a broad range of sorghum diversity, the sample size used in this study (306 accessions) is relatively small for GWAS, potentially limiting our power to detect genes with smaller effects. Additionally, the functions of many of the identified candidate genes are based on predicted domains and homology to genes in other species, and therefore require experimental validation in sorghum. The "control" treatment, though not directly inoculated, was subject to natural field infection, which could have introduced some confounding effects (Pourhoseingholi et al. 2012). Finally, the inherent complexity of ML models, while offering advantages in capturing complex interactions, can also make it more challenging to definitively pinpoint individual SNPs with the same level of statistical certainty as traditional GWAS. Future research should prioritize validating these findings in diverse environments and genetic backgrounds, as well as functionally characterizing the most promising candidate genes using techniques such as gene editing.

In conclusion, this study demonstrates the power of ML-driven GWAS to dissect the complex genetic architecture of polygenic disease resistance, providing a deeper understanding of plant-pathogen interactions and offering valuable resources for crop improvement. By leveraging ensemble methods and diverse phenotypic representations, we identified a suite of candidate genes associated with grain mold resistance in sorghum, highlighting the multifaceted nature of this trait and suggesting potential targets for enhancing durable resistance. These findings not only advance our understanding of sorghum's defense mechanisms but also provide a methodological framework applicable to the study of other complex traits in diverse organisms. The identified SNPs and candidate genes represent valuable resources for marker-assisted selection and genomic selection, ultimately contributing to the development of more resilient crop varieties and enhancing food security in regions challenged by this devastating disease. Further functional characterization of these genes promises to reveal fundamental insights into the intricate interplay between plants and their pathogens.

DATA AVAILABILITY

The complete list of top SNPs, their associated LD blocks, and the complete results of the Gene Ontology (GO) enrichment analysis are available in Supplementary Data 1. All other relevant data that support the findings of this study are available from the corresponding author upon reasonable request.

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The authors declare no competing interests.

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